```
import pandas as pd
import numpy as np

# PROBLEM 1: Create a Student DataFrame
data = {
    "Student": ["David", "Samuel", "Terry", "Evan"],
    "Age": [27, 24, 22, 32],
    "Country": ["UK", "Canada", "China", "USA"],
    "Course": ["Python", "Data Structures", "Machine Learning",
               "Web Development"],
    "Marks": [85, 72, 89, 76]
}

df = pd.DataFrame(data)
print("PROBLEM 1: Student DataFrame")
print(df)

# PROBLEM 2: Retrieve Marks column and assign to b
b = df["Marks"]
print("\nPROBLEM 2: Marks Column")
print(b)

# PROBLEM 3: Retrieve Country and Course columns and assign to c
c = df[["Country", "Course"]]
print("\nPROBLEM 3: Country and Course Columns")
print(c)

# PROBLEM 4: Data Loading and Exploration
customer_data = {
    "Customer_ID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Name": ["Aman", "Priya", "Rahul", "Neha", "Karan",
             "Simran", "Arjun", "Riya", "Vikram", "Anjali"],
    "Age": [25, 32, 28, 35, 40, 22, 31, 27, 45, 29],
    "Gender": ["Male", "Female", "Male", "Female", "Male",
               "Female", "Male", "Female", "Male", "Female"],
    "Purchase_Amount": [2000, 3500, 1500, 4000, 5000,
                        2500, 3000, 1800, 4500, 2200]
}

df = pd.DataFrame(customer_data)

print("\nPROBLEM 4: First Five Rows")
print(df.head())

print("\nSummary Statistics")
print(df.describe())

print("\nMissing Values")
```

```
print(df.isnull().sum())

# Handle missing values
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Gender"] = df["Gender"].fillna("Unknown")
df["Purchase_Amount"] = df["Purchase_Amount"].fillna(
    df["Purchase_Amount"].median()
)

print("\nDataset After Handling Missing Values")
print(df)

# PROBLEM 5: Data Filtering and Selection
print("\nPROBLEM 5: Customers Older Than 30")
print(df[df["Age"] > 30])

male_total = df.loc[
    df["Gender"] == "Male", "Purchase_Amount"
].sum()

female_total = df.loc[
    df["Gender"] == "Female", "Purchase_Amount"
].sum()

print("\nTotal Purchase Amount for Male Customers:", male_total)
print("Total Purchase Amount for Female Customers:", female_total)

selected = df[["Customer_ID", "Name", "Purchase_Amount"]]
print("\nSelected Customer Columns")
print(selected)

# PROBLEM 6: Data Transformation and Aggregation
# Apply a 10% discount
df["Discounted_Price"] = df["Purchase_Amount"] * 0.90

print("\nPROBLEM 6: Dataset with Discounted Prices")
print(df)

# Calculate average purchase amount by gender
average_purchase = df.groupby("Gender")["Purchase_Amount"].mean()
print("\nAverage Purchase Amount by Gender")
print(average_purchase)

# Sort purchases in descending order
sorted_df = df.sort_values(
    by="Purchase_Amount", ascending=False
)

print("\nDataset Sorted by Purchase Amount (Descending)")
print(sorted_df)
```